

Project: Dynamic Picking Replenishment (ZPA)

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System: SPEED WMS

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I. Summary

- **Goal:** Automatically replenish picking locations (ZPA) in multiples of VL cartons, filling to the location volume (location volume vs. article volume × quantity).
 - **Net requirement calculation:** portfolio requirements in units – available ZPA stock – ZPA replenishment pending (real quantities).
 - **Operator optimization:** chain missions from the same origin support via support scan.
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II. Context & Scope

- Dynamic picking replenishment happens in SPEED via the ZPA transfer.
- Standard system limitations require multiple RF steps for operational needs. The goal is to make dynamic location replenishment more efficient by fully automating the process.

- This process is **distinct from wave launches** but is driven by the order portfolio.

Scope:

- ZPA dynamic replenishment in multiples of VL cartons from reserve to dynamic picking locations.
- Priority: same article/lot.
- Execution of replenishment mission.

Target steps (hypotheses):

1. ZPA placement rule (find location by volume + deduct quantities from previous steps).
2. Picking replenishment module – optimize chaining of replenishments from the same origin location.

III. Definitions & Parameters

Term	Description
VL pallet (PCB palette)	Units per full pallet (e.g., 100)
VL carton (PCB carton)	Units per carton (configurable)
Unit volume (VU)	cm ³ per unit (configurable per article)
Location capacity	Maximum volume in cm ³ (configurable)
Replenishment rounding rule	Minimum = VL carton; always multiples of VL carton
Mixing lots in dynamic picking	Not allowed by default
Swelling coefficient	Allows placement even if within $\pm X\%$ of location volume

IV. Current Issue

- WMS proposes transferring the **full original support** to the dynamic picking location.
- If support volume exceeds location capacity → operator manually enters partial quantity to complete replenishment, then splits the original support and stores remnants manually elsewhere.

- No multi-location guidance, no consideration of subsequent manual placement destinations.

Consequences:

- RF overload, error risks, wasted time, stock inconsistencies.

V. Target Solution

- **Net requirement calculation (per article/lot):** Needs – ZPA available – pending ZPA replenishment.

Example:

Order	Cmd1	Cmd2	Cmd3	Total
Raw need (article A)	10	102	33	145
Picking units (raw)	10	2	33	45
ZPA unallocated stock	0	0	0	0
Pending ZPA replenishment	0	0	0	0
Net picking need	45			

- Round net need to VL carton: **45 → 50**
- **Target search:** first fill existing ZPA locations (same article/lot) to volume, in multiples of VL carton. If capacity < VL carton → skip and open new locations per placement rules.

Stepwise example:

Step	Location	Available volume	Possible cartons	Proposed	Deposited qty	Remaining
0	—	—	—	—	0	50
1	E1	96000	2.133	2	20	30
2	E2	96000	2.133	2	20	10
3	E3	96000	2.133	2	10	0

- **Mission chaining:** create 3 missions linked to same origin support with replenishment qty 20/20/5 → target locations E1, E2, E3.

- **Reservations & consistency:** consider these missions in next ZPA transfers.

Second launch example:

Order	Cmd4	Cmd2	Cmd3	Total
Raw need (article A)	25	102	33	160
Picking units (raw)	25	2	33	60
ZPA unallocated stock	30	—	—	—
Pending ZPA replenishment 5	—	—	—	—
Net picking need	25			

- Round net need: **25 → 30**

Stepwise example 2:

Step	Location	Available volume	Possible cartons	Proposed	Deposited qty	Remaining
0	—	—	—	—	0	30
1	E3	51000	1.133	1	10	20
2	E6	96000	2.133	2	20	0

- Consider pending missions even if not yet executed.

VI. Business Rules

- Must be multiples of VL cartons; never single units.
 - Strict capacity respect (volume) including swelling coefficient.
 - Priority: existing picking locations (same article/lot). No lot mixing.
 - Origin support selection: FEFO, DAEN, sufficient quantity.
 - Key parameters: location volume, unit volume, VL carton volume, swelling coefficient.
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VII. RF Ergonomics

Option 1:

- Scan origin support → display ZPA route (location 1...X & quantities).
- Guided to each target → deposit calculated qty → validation.
- Random adaptation: allow lower quantity.
- End of route: return to mission list. Remaining qty stays in reserve location.

Option 2:

- Scan origin → retrieve mission 1
- Go to target 1 → deposit qty → validation
- Automatically continue to mission 2 using same origin location
- Deposit qty → validation
- End of route → return to mission list, leftover stays in reserve

Option 3:

- Scan origin → retrieve mission 1
- Go to target 1 → deposit qty → validation
- Return to mission list
- Allow scan of support to enter mission instead of location (avoids going back to location)
- Deposit qty → validation
- Return to mission list

VIII. Points of Attention

- Deduct pending replenishments in **real quantities** (not origin support capacity).
 - Replenish in multiples of cartons; ZPA transfer already in VL carton multiples → check parameterization and impact.
 - Lot mixing not allowed by default.
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